PROJECT BRIEF · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 3: Project: Search Showdown

Module 1 | 50 minutes | Project: comparing algorithms | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Your mission
Run linear and binary search on the same sorted list of 100 numbers, count the steps each takes for several targets, and report which is more efficient — with each search in its own procedure.

Checklist

- [ ] Sorted list of 100
- [ ] Linear search with steps
- [ ] Binary search with steps
- [ ] Procedures
- [ ] Both reported
- [ ] Results table

Milestones

1. Data — done on: _____
2. Linear procedure — done on: _____
3. Binary procedure — done on: _____
4. Experiment — done on: _____
5. Conclusion and submit — done on: _____

Plan it

Sketch your stage and write what each sprite will do.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Algorithms	Neither works	One works	Both correct, including not found	Both correct and efficient
Measurement	None	Inconsistent counting	Fair, consistent step counts	Automated tests across many targets
Procedures	None	Partly	Each search in a procedure	Clean, reusable design
Results	None	One target	Table of 5 targets	Averages or graph
Conclusion	None	States a winner	Evidence-based conclusion	Explains when each algorithm is best

Want more?

- Automate 100 targets
- Average steps
- Bar chart with the pen

Two stars and a wish

Ask a friend to test your project.

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